Shipping Service Setup Guide — CentOS 9 (Updated)

This version follows the original Shipping CentOS 9 setup guide and incorporates the actual changes found during the Ansible deployment troubleshooting.

1. Install Maven

```
dnf install maven -y
```

2. Add Application User

```
useradd roboshop
```

3. Create Application Directory

```
mkdir /app
```

4. Download Application Code

```
curl -L -o /tmp/shipping.zip https://roboshop-builds.s3.amazonaws.com/shipping.zip
```

5. Extract Application Code

```
cd /app
unzip /tmp/shipping.zip
```

6. Build the Application

```
cd /app
mvn clean package
```

7. Rename the JAR File — CHANGE REQUIRED

The original guide uses **target/shipping-1.0.jar**. During the actual deployment, Maven generated **/app/target/shipping.jar**. Therefore the Ansible command must use the actual generated filename:

```
mv target/shipping.jar shipping.jar
```

This produces **/app/shipping.jar**, which matches the systemd ExecStart path.

8. Configure SystemD Shipping Service

Keep the service structure from the reference guide. The important path is /app/shipping.jar. Replace the Cart and MySQL placeholders with the actual server addresses.

```
[Unit]
Description=Shipping Service

[Service]
User=roboshop
Environment=CART_ENDPOINT=<CART-SERVER-IPADDRESS>:8080
Environment=DB_HOST=<MYSQL-SERVER-IPADDRESS>
ExecStart=/bin/java -jar /app/shipping.jar
SyslogIdentifier=shipping

[Install]
WantedBy=multi-user.target
```

9. Reload SystemD

```
systemctl daemon-reload
```

10. Enable and Start Shipping Service

```
systemctl enable shipping
systemctl start shipping
```

11. Install MySQL Client

```
dnf install mysql -y
```

12. Load Shipping Database Schema — CHANGE REQUIRED

The reference guide points to **/app/schema/shipping.sql**, but that file did not exist on the deployed application. The actual application contained these files under **/app/db/**:

```
/app/db/
■■■ app-user.sql
■■■ master-data.sql
■■■ schema.sql
```

The actual schema file creates the **cities** database and cities table, so the schema command was changed to:

```
mysql -h mysql.roboshopservice.store -uroot -pRoboShop@1 < /app/db/schema.sql
```

13. Create Shipping Database User — ADDED

The deployed application also contains **/app/db/app-user.sql**, which creates the shipping database user and grants access to cities.*.

```
mysql -h mysql.roboshopservice.store -uroot -pRoboShop@1 < /app/db/app-user.sql
```

14. Load Master Data — ADDED

The deployed application contains **/app/db/master-data.sql** (about 64 MB). If the lab requires the city master data, load it after the schema:

```
mysql -h mysql.roboshopservice.store -uroot -pRoboShop@1 < /app/db/master-data.sql
```

15. Restart Shipping Service

```
systemctl restart shipping
```

16. Verify Shipping Service

```
systemctl status shipping
ss -lntp
curl http://localhost:8080/codes
```

Troubleshooting Changes Identified

Reference guide	Actual deployment	Required change
target/shipping-1.0.jar	target/shipping.jar	Use mv target/shipping.jar shipping.jar
/app/schema/shipping.sql	/app/db/schema.sql	Load /app/db/schema.sql
No separate user step	/app/db/app-user.sql exists	Load app-user.sql
No master-data step shown	/app/db/master-data.sql exists	Load master-data.sql when required

Important: The original guide's CART_ENDPOINT and DB_HOST placeholders still need to be replaced with the actual Cart and MySQL server addresses. The changes above are limited to the differences confirmed during the deployment troubleshooting.

Final Command Sequence Used for the Updated Deployment

```
dnf install maven -y
useradd roboshop
mkdir /app
curl -L -o /tmp/shipping.zip https://roboshop-builds.s3.amazonaws.com/shipping.zip
cd /app
unzip /tmp/shipping.zip
mvn clean package
mv target/shipping.jar shipping.jar

# create /etc/systemd/system/shipping.service
systemctl daemon-reload
systemctl enable shipping
systemctl start shipping

dnf install mysql -y
mysql -h mysql.roboshopservice.store -uroot -pRoboShop@1 < /app/db/schema.sql
mysql -h mysql.roboshopservice.store -uroot -pRoboShop@1 < /app/db/app-user.sql
mysql -h mysql.roboshopservice.store -uroot -pRoboShop@1 < /app/db/master-data.sql

systemctl restart shipping
systemctl status shipping
```